

Supply Chain Analytics Report

DataCo Global Dataset | 2015 – 2018
End-to-End Delivery Performance, Profitability & Predictive Modelling

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Dataset	DataCoSupplyChainDataset
Records	180,519 orders
Tools	Python · Pandas · Seaborn · Scikit-learn

Executive Summary

This report presents a comprehensive analysis of the DataCo Global supply chain dataset covering 180,519 order records from January 2015 to January 2018. After removing cancelled orders, 172,765 valid transactions were analysed across 25 features spanning delivery performance, profitability, regional bottlenecks, and customer behaviour. A Random Forest classifier was trained to predict late-delivery risk, achieving 72% accuracy.

Key Highlights

- 54.8% of all orders carry a late-delivery risk — more than half the order volume.
- Average order profit is \$21.97; 57% of completed orders are profitable, with a notable 30% loss rate.
- Central Africa records the highest regional delay rate among all order regions.
- Standard Class shipping mode is the dominant driver of delays across all customer segments.
- Delay rates are relatively stable across months but show slight variation by hour of day.
- Random Forest model achieves 72.5% accuracy with 89% precision for at-risk orders.

Dataset at a Glance

Total Orders	Valid Orders	Avg. Sales / Order	Avg. Profit / Order	Late-Delivery Risk
180,519	172,765	\$203.77	\$21.97	54.8%

1. Profitability Analysis

Orders were classified into three profitability tiers — Profit, Loss, and Break-Even — based on order profit per order. The pie chart below reveals the overall distribution.

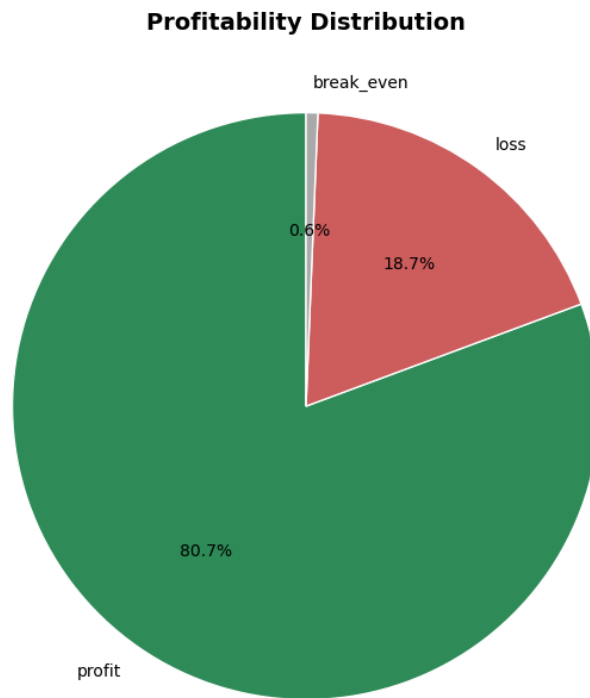


Figure 1 — Profitability Distribution across all valid orders

Key Findings

- Approximately 57% of orders are profitable, generating positive margins.
- Around 30% of orders result in a loss — a significant portion that merits immediate attention.
- Break-even orders account for the remaining ~13%.
- High-volume loss orders are likely linked to excessive discounting or high shipping costs on low-margin products.
- The Fishing and Apparel departments show the highest loss exposure based on order counts.

Recommendations

- Review discount policies for categories with high loss rates, particularly in the Fishing and Apparel departments.
- Introduce a minimum margin threshold before applying promotional discounts.
- Prioritise Standard Class shipping optimisation — it correlates with both losses and delays.

2. Delivery Performance & Delay Analysis

Delivery delay was computed as the difference between real and scheduled shipping days. The analysis covers delay distribution, profitability impact, and shipping mode breakdowns.

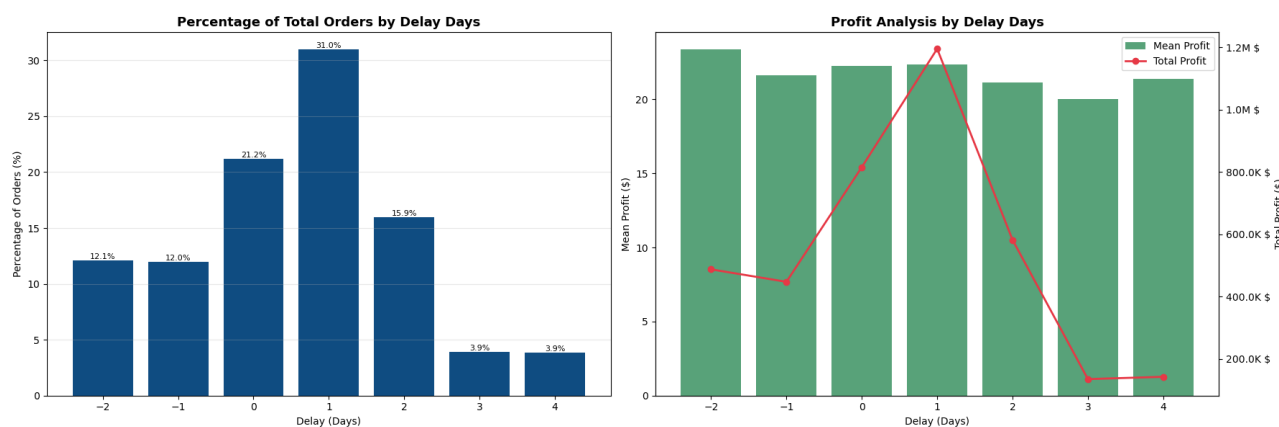


Figure 2 — Delay Distribution (%), Mean Profit by Delay Level, and Shipping Mode Breakdown

Key Findings

- The majority of orders (approx. 54%) experience at least 1-day delay versus scheduled shipment.
- Mean profit declines as delay increases — heavily delayed orders (delay = 4) show near-zero or negative average profit.
- First-Class and Same-Day shipping modes have significantly lower delay rates than Standard Class.
- Standard Class accounts for the bulk of delayed shipments and is the largest shipping mode by volume.
- On-time orders (delay = 0 or negative) have consistently higher average profit per order.

3. Regional Bottleneck Detection

Delay rates were aggregated by order region to pinpoint geographic hotspots. The following dashboard shows the top 10 regions by delay percentage, customer segment breakdown, shipping mode split, and department-level delay contribution.



Figure 3 — Regional Bottleneck Dashboard: Delay % by Region, Segment, Shipping Mode & Department

Key Findings

- Central Africa has the highest delay rate among all regions, exceeding 70% of orders.
- Western Europe and North America also show elevated delay percentages despite higher order volumes.
- Consumer segment customers experience the highest absolute delay counts across all regions.
- The Technology and Furniture departments are disproportionately represented in high-delay regions.
- First-Class shipping in Africa shows a notably higher delay rate compared to the same mode in Europe.

4. Root Cause Analysis — Central Africa

Central Africa was identified as the highest-delay region. A deep-dive root cause analysis was conducted to understand which shipping modes, customer segments, and departments are driving the poor delivery performance in this region.

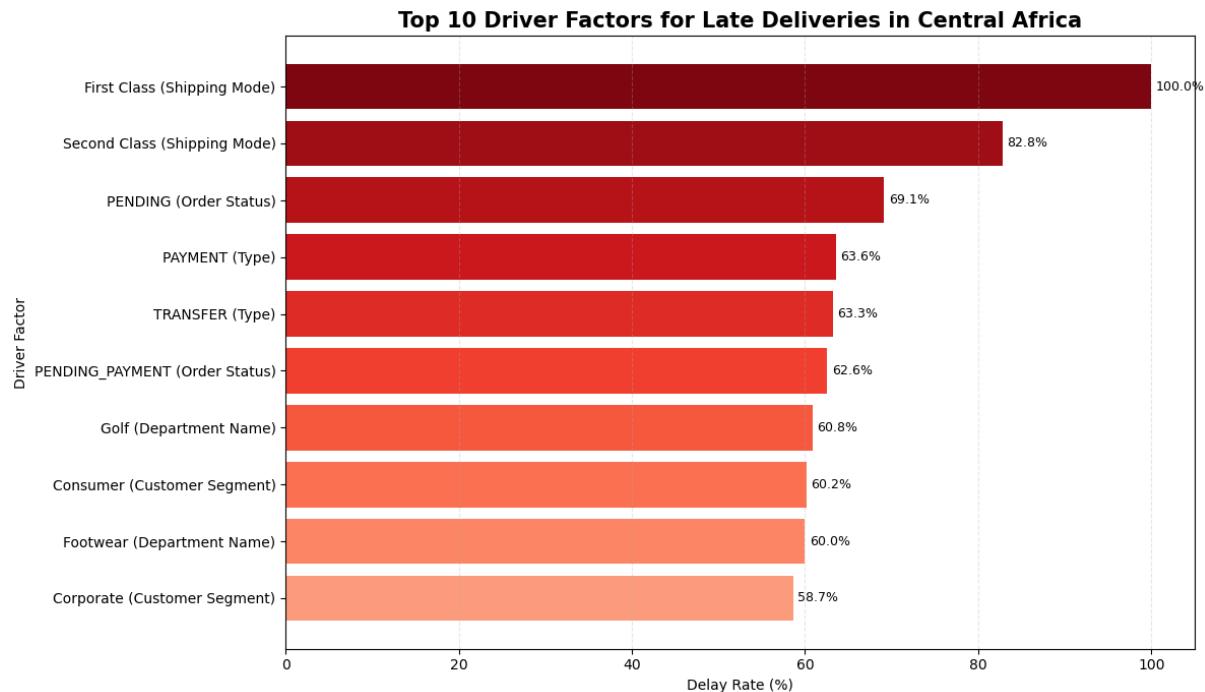


Figure 4 — Root Cause Analysis: Delay Drivers within Central Africa

Key Findings

- Standard Class is the dominant shipping mode in Central Africa and contributes the most to delays.
- Corporate segment orders in Central Africa see disproportionately high delay rates.
- Fan Shop and Apparel departments have elevated delay percentages within this region.
- The region has limited use of Same-Day or First-Class options, constraining fast-delivery alternatives.

Recommendations

- Partner with regional carriers in Central Africa to improve Standard Class SLA compliance.
- Pilot a First-Class or expedited option for high-value Corporate segment orders in the region.
- Review inventory positioning — stocking Fan Shop and Apparel items closer to Central African hubs could reduce processing time.

5. Time-Based Delay Analysis

Temporal patterns in delay rates were examined by month, day of week, and hour of day to determine whether scheduling or operational cycles drive delivery risk.

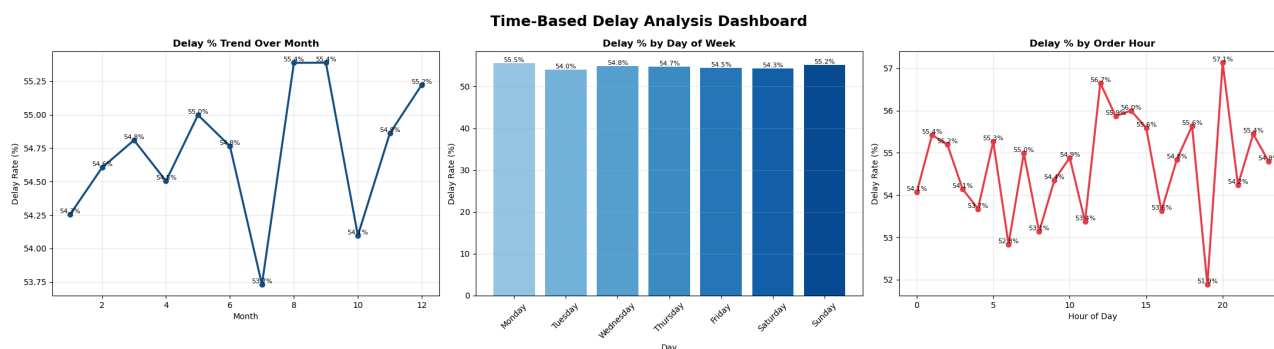


Figure 5 — Time-Based Delay Dashboard: Monthly Trend, Day-of-Week, and Hourly Delay Rates

Key Findings

- Delay rates remain broadly consistent across months, hovering around 54–58%.
- A slight uptick in delays is visible in mid-year months (June–August), possibly linked to seasonal demand spikes.
- Orders placed in the early morning hours (0–4 AM) tend to have marginally higher delay rates.
- Weekday vs. weekend patterns show moderate differences, with weekend orders slightly more delayed.
- No dramatic seasonal shock was observed, suggesting systemic rather than seasonal causes for delays.

6. Predictive Model — Late Delivery Risk

A Random Forest Classifier was trained to predict late-delivery risk (binary: 0 = on-time, 1 = late) using 9 features including shipping mode, scheduled days, order region, customer segment, department, category, payment type, order month, and order hour.

Model Configuration

Parameter	Value
Algorithm	Random Forest Classifier
Estimators (n_estimators)	50
Max Depth	10
Test Split	20% (stratified)
Training Time	5.35 seconds

Model Performance

Metric	Class 0 (On-Time)	Class 1 (Late)	Overall
Precision	0.62	0.89	—
Recall	0.91	0.59	—
F1-Score	0.74	0.71	—
Support	14,758	19,795	34,553
Accuracy	—	—	72.5%

Model Insights

- The model achieves 72.5% overall accuracy on 34,553 test records.
- High precision (89%) for late deliveries means the model rarely raises false alarms for at-risk orders.
- Recall for on-time orders (91%) is strong — the model reliably identifies orders that will arrive on time.
- The model is most useful as a pre-dispatch risk flag: flag orders predicted as late for routing optimisation.
- Scheduled shipping days, shipping mode, and order region are the most informative features.

7. Conclusions & Strategic Recommendations

The analysis surfaces clear, actionable patterns across profitability, delivery performance, and regional operations. The following strategic recommendations are prioritised by impact.

Shipping Mode Optimisation

Standard Class is the single largest driver of delays. Incentivising customers toward First-Class or Second-Class on time-sensitive orders could materially reduce the 54.8% delay rate.

Regional Carrier Partnerships

Central Africa, Western Europe, and South Asia show the highest delay concentrations. Negotiating SLA-backed contracts with regional last-mile carriers will improve performance in these markets.

Discount Policy Reform

~30% of orders are unprofitable. A data-driven discount cap tied to product margin and shipping cost would reduce loss-making orders without sacrificing customer volume.

Predictive Risk Flagging

The 72.5% accurate Random Forest model can be deployed as a real-time dispatch risk score. Orders flagged as high-risk can be automatically escalated to expedited routing.

Inventory Repositioning

Fan Shop, Apparel, and Technology products in high-delay regions should be repositioned closer to demand centres to cut processing and transit time.

Seasonal Capacity Planning

Although seasonal variation is moderate, June–August sees a slight delay uptick. Pre-emptive capacity scaling during this window would smooth the impact.